Fundamental Theorem of Algebra, Multiple Root Theorem

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Based on the lecture note uploaded on SPeCTRUM

Sem I, 2025/2026

Learning Outcomes

At the end of this lecture, you should be able to:

□ Understand and apply the Fundamental Theorem of Algebra.

□ Understand what is the multiplicity of a root.

□ Find the derivative of polynomials.

□ Understand and apply the Multiple Root Theorem.

□ Know about the conjugate roots in real polynomials and know how to use this result.

□ Understand and apply the Rational Roots Theorem.

Fundamental Theorem of Algebra:
Every polynomial of degree $n > 0$ has exactly n roots in $\mathbb{C}$, counting multiplicities.

not necessarily distinct.

Every polynomial $P(x)$ with degree $n > 0$ can be factorised as follows:

degree n polynomial.

$$P(x) = c(x - \alpha_1)(x - \alpha_2)\dots(x - \alpha_n)$$

Factorisation theorem.

leading coefficient. the coefficient of $x^n = c$

for some complex numbers $c, \alpha_1, \alpha_2, \dots, \alpha_n$.
These $\alpha_1, \alpha_2, \dots, \alpha_n$ are the roots of $P(x)$ and they are not necessarily distinct.

$P(\alpha_1) = c\underbrace{(\alpha_1 - \alpha_1)}_{=0}(\alpha_1 - \alpha_2)\dots(\alpha_1 - \alpha_n) = 0$

Example:
$p(x) = x^4 - 1 = (x^2 + 1)(x^2 - 1) = (x + i)(x - i)(x + 1)(x - 1)$ has 2 real roots -1 and 1, but it has 4 complex roots $-i$, i, -1, and 1.

Example:

Let $P(x) = 2x^3 - 3x^2 + kx + 3$, where k is a constant. Find the value of k if $-\frac{1}{2}$ is a root of $P(x)$ and then obtain all the other roots of $P(x)$.

$P(-\frac{1}{2}) = 0$

$P(-\frac{1}{2}) = 0$

$2(\frac{1}{2})^3 - 3(\frac{1}{2})^2 + k(\frac{1}{2}) + 3 = 0$

$k = 4.$

$P(x) = 2x^3 - 3x^2 + 4x + 3.$

$= (x+\frac{1}{2})(2x^2 - 4x + 6)$ polynomial of degree 2.

$P(x) = 0 \Rightarrow x = -\frac{1}{2}$ or $2x^2 - 4x + 6 = 0$

$\Rightarrow x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(2)(6)}}{2(2)}$

$= \frac{4 \pm \sqrt{-32}}{4}$

$$
\begin{array}{r}
2x^2 - 4x + 6 \\
x + \frac{1}{2} \overline{\big) 2x^3 - 3x^2 + 4x + 3} \\
(-) \quad 2x^3 + x^2 \qquad\qquad \\
\hline
-4x^2 + 4x + 3 \\
(-) \quad -4x^2 - 2x \qquad \\
\hline
6x + 3 \\
6x + 3 \\
\hline
0
\end{array}
$$

$$= \frac{4 \pm 4\sqrt{2}\, i}{4}$$

$$= \underline{1 \pm \sqrt{2}\, i}$$

$x = 1 + \sqrt{2}i$, $x = 1 - \sqrt{2}i$

Hence, the roots of $P(x) = 0$ are $-\frac{1}{2}$, $1 + \sqrt{2}i$, $1 - \sqrt{2}i$

Example:
Find the polynomial $P(x)$ of degree 3 such that $P(1) = P(-2) = 0$, $P(3) = 130$ and $P(-3) = -20$.

Since $P(1) = 0$, $P(-2) = 0$, it follows that $(x-1)(x+2)$ are the factors of $P(x)$

So, we can write $P(x) = (x-1)(x+2)(ax+b)$ ← degree 1

we want to find a & b.

$P(3) = (\cancel{3}-1)(\cancel{3}+2)(3a+b) = 130$ (2, 5)

$3a + b = 13$ — ①

$P(-3) = (-3-1)(-3+2)(-3a+b) = -20$ (−4, −1)

$-3a + b = -5$ — ②

① - ②, $6a = 18 \Rightarrow a = 3$

From ①, $3(3) + b = 13 \Rightarrow b = 4$

Hence, $P(x) = (x-1)(x+2)(3x+4)$

$= 3x^3 + 7x^2 - 2x - 8$

Root of Multiplicity k

Let k be a positive integer. A number a is said to be a root of multiplicity k of $P(x)$ if

1. $(x-a)^k$ is a factor of $P(x)$, and
2. $(x-a)^{k+1}$ is not a factor of $P(x)$.

A root of multiplicity two is called a double root.

Let c be a root of multiplicity k of a polynomial $P(x)$.

- If $k = 1$, then c is called a simple root.
- If $k > 1$, then c is called a multiple root.

Example:
Suppose that $p(x) = (x-1)^3(x+2)(x-i)^2$. Then 1 is a root of multiplicity 3; -2 is a root of multiplicity 1; i is a root of multiplicity 2.

Derivatives of a Polynomial

Let $f(x) = a_0 + a_1x + a_2x^2 + \overset{a_3x^3}{\cdots} + a_nx^n = \sum_{i=0}^{n} a_ix^i$ be a polynomial.

The first derivative of $f(x)$ is

$$f'(x) = a_1 + 2a_2x + 3a_3x^2 + \overset{4a_4x^3}{\cdots} + na_nx^{n-1} = \sum_{i=1}^{n} ia_ix^{i-1}$$
$$= \sum_{i=0}^{n} ia_ix^{i-1}$$

The second derivative of $f(x)$ is the derivative of $f'(x)$, that is

$$f''(x) = 2a_2 + 2(3)a_3x + (3)(4)a_4x^2 + \dots + (n-1)(n)a_nx^{n-2}$$
$$= \sum_{i=2}^{n} (i-1)ia_ix^{i-2}$$

Continue the process, we define the k^{th} derivative of $f(x)$ for $k > 2$ as

$$f^{(k)}(x) = \sum_{i=k}^{n} i(i-1)(i-2)\dots(i-k+1)a_ix^{i-k}$$

$$f^{(k)}(x) = 0, \quad \forall k > n$$

$$f^{(n)}(x) = n(n-1)(n-2)\dots(3)(2)(1)a_n = (n!)a_n$$

Derivatives of a Polynomial

Let $f(x) = 3 + 4x + 5x^2 + 6x^4$.

$f'(x) = 4 + 10x + 24x^3$

$f''(x) = 10 + 72x^2$

$f'''(x) = 144x$

$f^{(4)}(x) = 144 \leftarrow 4! \times 6$

$f^{(5)}(x) = 0$

$f^{(k)}(x) = 0, \ \forall k \geq 5$

For polynomials $f(x)$ and $g(x)$,

- If $p(x) = f(x) + g(x)$, then $p'(x) = f'(x) + g'(x)$;
- If $p(x) = f(x)g(x)$, then $p'(x) = f'(x)g(x) + f(x)g'(x)$; (Product rule)
- If $p(x) = cf(x)$ where c is a constant, then $p'(x) = cf'(x)$;
- If $p(x) = c(x-a)^m$, $m \geq 1$, then $p'(x) = cm(x-a)^{m-1}$

Multiple Root Theorem

Suppose that a is a root of multiplicity 2 of polynomial $p(x)$.

Then $p(x) = (x-a)^2 q(x)$ where $x-a$ is not a factor of $q(x)$

$(x-a)^3$ is not a factor of $p(x)$

So, $p'(x) = 2(x-a)q(x) + (x-a)^2 q'(x)$

$\Rightarrow p'(a) = 0 + 0 = 0$

$p''(x) = 2q(x) + 2(x-a)q'(x) + 2(x-a)q'(x) + (x-a)^2 q''(x)$

$\Rightarrow p''(a) = 2q(a) \neq 0$ $\quad q(a) \neq 0$ because $x-a$ is not a factor of $q(x)$

a is a root of $p(x)$ with multiplicity 2

$\Rightarrow p(a) = 0, \ p'(a) = 0, \ p''(a) \neq 0$.

Multiple Root Theorem:
Let k be a positive integer. Then a is a root of multiplicity k of a polynomial $P(x)$ if and only if

$$P(a) = P'(a) = \cdots = P^{(k-1)}(a) = 0 \text{ and } P^{(k)}(a) \neq 0.$$

Example:
Find all the roots of $P(x) = x^4 - 6x^2 + 8x - 3$ if one of them is of multiplicity three.

$p(x) = x^4 - 6x^2 + 8x - 3$

$p'(x) = 4x^3 - 12x + 8$

$p''(x) = 12x^2 - 12$

$p'''(x) = 24x$

Let a be the root of $p(x)$ with multiplicity 3.

$p''(a) = 0 \Rightarrow 12a^2 - 12 = 0$

$a = \pm 1$

$p(-1) = (-1)^4 - 6(-1)^2 + 8(-1) - 3 \neq 0$

$1 \quad -6 \quad -8 \quad -3$

$p(1) = (1)^4 - 6(1)^2 + 8(1) - 3 = 0$

$p'(1) = 4(1)^3 - 12(1) + 8 = 0$

So 1 is the root of $p(x)$ with multiplicity 3.

$(x-1)^3$ is a factor of $p(x)$ and $p(x) = (x-1)^3(cx+d)$

$= x^4 - 6x^2 + 8x - 3$

$= (x-1)^3(x-3)$

$(x-1)^3(cx-d) = x^4 - 6x^2 + 8x - 3$

Comparing the coefficient of x^4 : $c = 1$

Comparing the coefficient of x^0 : $d = -3$

The roots are:

1, 1, 1, 3

Complex Roots

If $z = a + bi$ $(a, b \in \mathbb{R}, b \neq 0)$ is a root of the real polynomial $P(x)$, then its conjugate $a - bi$ is also a root of $P(x)$.

↳ all coefficients of $P(x)$ are real

Proof

z is a root of $P(x)$ $\Rightarrow$ $x - z$ is a factor of $P(x)$.

Now, we wish to show that $x - \bar{z}$ is also a factor of $P(x)$.

$$\text{Let } N(x) = (x - z)(x - \bar{z}) = x^2 - \bar{z}x - zx + z\bar{z}$$
$$= x^2 - (z + \bar{z})x + z\bar{z}$$
$$= x^2 - [(a+bi) + (a-bi)]x + (a+bi)(a-bi)$$
$$= x^2 - 2ax + (a^2 + b^2)$$

So, $N(x)$ is a real polynomial of degree 2.

When $P(x)$ is divided by $N(x)$, by division algorithm,

$$P(x) = q(x)N(x) + r(x) \quad , \quad \deg r(x) < \deg N(x) = 2$$

$$\deg r(x) \leq 1$$

$$= q(x)\,N(x) + cx + d$$

$$= q(x)\,(x-z)\,(x-\bar{z}) + cx + d.$$

Since z is a root of $P(x)$, $P(z) = 0$

$$cz + d = 0$$

$$c(a+bi) + d = 0$$

$$(ca+d) + (cb)i = 0$$

$$\Rightarrow cb = 0 \quad \text{and} \quad ca + d = 0$$

$$\Rightarrow d = 0$$

Since $b \neq 0$, $c = 0$

So, $P(x) = (x-z)(x-\bar{z})\,q(x)$ and $(x-\bar{z})$ is also a factor of $P(x)$ and $\bar{z}$ is a root of $P(x)$.

Example:
The polynomial $P(x) = (x - i)(x + 2) = x^2 + (2 - i)x - 2i$ is not a real polynomial. It has a complex root i, but the conjugate $-i$ is not a root of $P(x)$.

$P(i) = (i - i)(i + 2) = 0$

$P(-i) = (-i - i)(-i + 2) = -2i(-i + 2)$
$= -2 - 2i \neq 0$

Example:
Find all roots of the equation $2x^3 - 7x^2 + 10x - 6 = 0$ if one of its roots is $1 + i$.

Let $p(x) = 2x^3 - 7x^2 + 10x - 6 = 0$. Since $p(x)$ is a real polynomial and $1+i$ is a root of $p(x)$, the conjugate $1-i$ must also be a root of $p(x)$.

So $(x - (1+i))(x - (1-i)) = x^2 - 2x + 2$ is a factor of $p(x)$

Then, $p(x) = (x^2 - 2x + 2)\, q(x) \Leftarrow$ find $q(x)$

By doing the long division ← Exercise.

$$p(x) = (x^2 - 2x + 2)(2x - 3)$$

$$= (x - (1+i))(x - (1-i))(2x - 3)$$

So $p(x)$ has 3 roots $1+i$, $1-i$, $\frac{3}{2}$.

A rational number $\frac{p}{q}$ $(p, q \in \mathbb{Z}, q \neq 0)$ is said to be in lowest term if p, q are relatively prime i.e., p and q have no common factors except ± 1.
(Eg. $\frac{3}{5}$ is in lowest term, but $\frac{6}{10}$ is not in lowest term.)

Rational roots theorem:
Let $f(x) = a_0 + a_1 x + \cdots + a_{n-1}x^{n-1} + a_n x^n$ be a polynomial with coefficients which are integers. If the rational number $\frac{p}{q}$ is a root of $f(x) = 0$ in lowest term, then $p | a_0$ and $q | a_n$, i.e., $a_0 = ps$ and $a_n = qt$ for some $s, t \in \mathbb{Z}$.

proof:

$\frac{p}{q}$ is a root of $f(x) \Rightarrow f\left(\frac{p}{q}\right) = 0$

$$a_0 + a_1\left(\frac{p}{q}\right) + a_2\left(\frac{p}{q}\right)^2 + \ldots + a_{n-1}\left(\frac{p}{q}\right)^{n-1} + a_n\left(\frac{p}{q}\right)^n = 0$$

(*) Multiply by q^n : $\underline{a_0 q^n} + a_1 p q^{n-1} + a_2 p^2 q^{n-2} + \ldots + a_{n-1} p^{n-1} q + a_n p^n = 0$

$$a_1 p q^{n-1} + a_2 p^2 q^{n-2} + \ldots + a_{n-1} p^{n-1} q + a_n p^n = -a_0 q^n$$

$$\underset{\text{integer.}}{p}\underbrace{(a_1q^{n-1}+a_2pq^{n-2}+\dots+a_{n-1}p^{n-2}q+a_np^{n-1})}_{\text{integer}} = \underbrace{-a_0q^n}_{\text{integer.}}$$

$\Rightarrow p \mid a_0q^n$

Since p and q are relatively prime, so $p \nmid q$,

so $p \mid a_0$

Now, we factoring out q from $(*)$

$$\underset{\text{integer}}{q}\underbrace{(a_0q^{n-1}+a_1pq^{n-2}+a_2p^2q^{n-3}+\dots+a_{n-1}p^{n-1})}_{\text{integer}} = \underbrace{-a_np^n}_{\text{integer}}$$

$\Rightarrow q \mid -a_np^n$

Since p and q are relatively prime, so it follows that

$q \mid a_n$.

Example:
Solve the equation $x^4 - x^3 - 3x^2 + 4x - 4 = 0$. ← find the roots.

Let $f(x) = x^4 - x^3 - 3x^2 + 4x - 4 = 0$

and let $\frac{p}{q}$ be some rational number in the lowest term.

By the Rational Root Theorem, if $\frac{p}{q}$ is a root of $f(x)$,

then $p | -4$ and $q | 1$ and $p = \pm 1, \pm 2, \pm 4$, $q = \pm 1$

The possible rational roots of $f(x)$ are $\pm 1, \pm 2, \pm 4$

Now we calculate $f(1) = -3 \neq 0$, $f(-1) = -9 \neq 0$

$f(2) = 0$, $f(-2) = 0$

$f(4) = 156 \neq 0$, $f(-4) = 252 \neq 0$

So 2 and -2 are the roots of $f(x)$ and $(x+2)$ and $(x-2)$ are the factors of $f(x)$

So $f(x) = (x+2)(x-2)\, q(x)$

By long division / synthetic division ← Exercise.

$$f(x) = (x+2)(x-2)(x^2 - x + 1)$$

By using the quadratic formula,

$x^2 - x + 1 = 0$ has the roots $\dfrac{-1 \pm \sqrt{3}\, i}{2}$

So $f(x) = 0$ has 4 roots, $-2, 2, -\frac{1}{2} - \frac{\sqrt{3}}{2} i, -\frac{1}{2} + \frac{\sqrt{3}}{2} i$

Rational Roots

Example:
Show that $3x^2 - x + 5 = 0$ has no rational roots.

Let $P(x) = 3x^2 - x + 5 = 0$

If $P(x)$ has a rational root $\frac{p}{q}$ in the lowest term,

$p \mid 5$ and $q \mid 3$

So, $p = \pm 1, \pm 5$ and $q = \pm 1, \pm 3$.

So the possible rational roots of $P(x)$ are $\pm 1, \pm \frac{1}{3}, \pm 5, \pm \frac{5}{3}$

Go and verify that $P(-1)$, $P(1)$, $P(-\frac{1}{3})$, $P(\frac{1}{3})$, $P(-5)$, $P(5)$

$P(-\frac{5}{3})$, $P(\frac{5}{3})$ are all not zero.

So $P(x) = 0$ has no rational root.